Targeted Repair of a Poisoned RetinaNet under a Hidden Clean Teacher

An evidence-based Kaggle study with architecture, EDA, training ablations and leaderboard forensics

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Abstract

The Neural Debris Removal competition required repair of a poisoned single-class RetinaNet while the clean teacher, clean training data and all test labels remained hidden. The available evidence comprised a 145.5 MB checkpoint, 20 annotated poison images, 2,000 unlabeled 1024 x 1024 16-bit images and the poisoned model's own detections. We reconstruct the full research trajectory from 40 authenticated Kaggle kernels, 19 scored submissions, checkpoint audits and post-competition solution reports. Unlike our earlier retrospective report, this study gives an explicit EDA, the detector architecture, the exact trainable scope and objective, hyperparameter sensitivity, inference equations and ablations.

The strongest private model repaired only the classification subnet: activation-guided pruning zeroed the top 15% poison-selective channels in each classification convolution, then 2,376,455 parameters (6.55% of the detector) were fine-tuned for 10 iterations on empty labels at learning rate 2.5e-4 with an EWC penalty of 250 toward the post-pruning weights. The backbone, feature pyramid and box-regression head remained frozen. At inference, original-model boxes were retained and their confidence was remapped from repaired-model survival and poison-box geometry. This candidate scored 229.1934 public and 306.3502 private, the best private result on the account. A later 219.4257 public entry did not train a model; it replayed a fixed 450-detection mask and regressed to 308.0971 private. The evidence therefore supports constrained classifier repair and geometry preservation, but not public-leaderboard mask optimization.

Keywords: machine unlearning, backdoor removal, object detection, RetinaNet, Detectron2, activation pruning, EWC, confidence calibration, Kaggle.

1. Introduction

Model repair differs from ordinary fine-tuning. The objective is not to learn a new detector from labels, but to remove a narrow poisoned behavior while retaining an unknown clean function. In this challenge the participant observes only the corrupted checkpoint and examples of the behavior to forget. Any weight update can reduce poison response, but the same update can erase real streak detections, move boxes, create unmatched false positives or distort confidence. The hidden clean teacher makes this preservation problem the dominant source of uncertainty.

Our first report documented the chronology, but it left three questions underdeveloped: what the images and poison boxes actually look like statistically; which modules of the fundamental detector were changed; and why the best model-based notebook generalized better than more aggressive alternatives. This revision answers those questions with measured evidence rather than leaderboard narrative.

Contributions

- An EDA of all 20 annotated poison images, a systematic 40-image test sample and all 2,593 poisoned-template detections.
- A tensor-level reconstruction of the ResNet-50-FPN RetinaNet, including anchor geometry and the exact 2.376 M parameter repair scope.

- A complete description of activation-guided pruning, classifier-only empty-label training, EWC anchoring and metric-aware confidence remapping.
- Ablations that separate pruning, short fine-tuning and public-score mask selection, including negative results.
- A forensic distinction between the best private model and the best public notebook, preventing a misleading claim of model improvement.

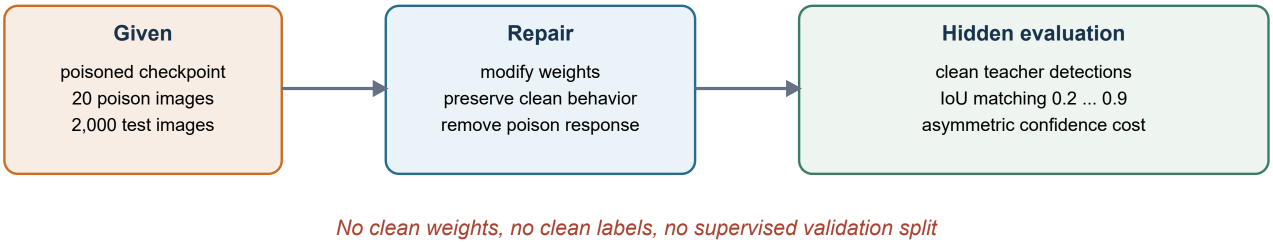


Figure 1. The challenge is constrained model repair against a hidden clean teacher, not ordinary supervised detection.

2. Challenge formulation and evaluation

2.1 Available information

Asset	Observed content	Role in the study
Poisoned checkpoint	301 tensors; 36.25 M parameters/buffers; SHA-256 f6c21f...657c	Starting detector and frozen reference
Unlearn set	20 uint16 PNGs; one annotated poison box per image	Defines behavior to forget
Test set	2,000 unlabeled uint16 PNGs	Permitted label-free diagnostics and final inference
Sample submission	2,593 detections; 634 empty images	Poisoned-model geometry and confidence baseline
Leaderboard	19 scored submissions; private split final	Sparse external evidence

2.2 maCADD

At IoU thresholds t in {0.2, 0.3, ..., 0.9}, clean and submitted detections are matched. Unmatched clean or submitted boxes incur their full confidence. For a matched pair, a confidence change in the intended unlearning direction is discounted by A = 10; a change in the wrong direction is charged fully. The competition score is an IoU-weighted mean and lower is better.

```
maCADD = sum_t [ t * ACADD(t) ] / sum_t t
```

This metric makes three design choices consequential. First, deleting a real detection is expensive because it becomes unmatched. Second, moving a box can be worse than modestly changing its confidence because matching can fail at high IoU. Third, confidence calibration is part of the task, not a cosmetic post-processing step. These facts motivate freezing the box regressor and using the original checkpoint as the candidate generator.

2.3 Evidence protocol

Claims are assigned one of three evidence levels. **Scored** means a Kaggle public/private value exists. **Audited no-submit** means code, hashes and output statistics exist but no leaderboard score is imputed. **Retrospective inference** means a mechanism is inferred from matched outputs and is labeled accordingly. The EDA uses all 20 unlearn images and a systematic, unlabeled 40-image test sample with IDs 0, 50, ..., 1950; it is not presented as a random population estimate.

3. Exploratory data analysis

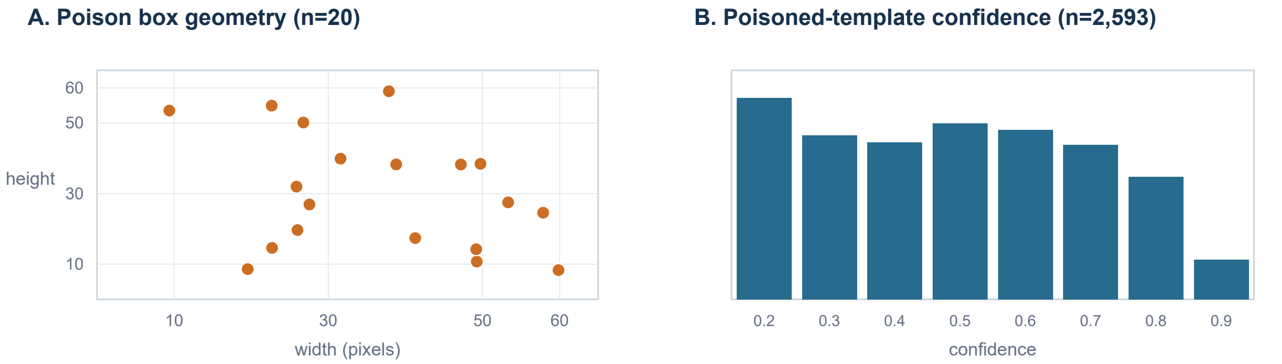
3.1 Image statistics

All images are 1024 x 1024, 16-bit grayscale PNGs. The loader divides uint16 values by 65,535, scales to 0..255 and repeats the channel three times to satisfy the COCO RetinaNet interface. Across the 20 unlearn images, median normalized intensity was 0.16784 and median per-image standard deviation was 0.07768. In the 40-image systematic test sample the corresponding values were 0.16798 and 0.07808. Median 1st-to-99th percentile dynamic range was 0.31405 versus 0.31377. Thus the unlearn set is not separated from test by global brightness, noise or dynamic range.

Per-image statistic	Unlearn, n=20	Test sample, n=40	Interpretation
Mean intensity, median	0.177713	0.177771	negligible global shift
Median intensity, median	0.167842	0.167979	same background level
Standard deviation, median	0.077681	0.078075	same noise/structure scale
MAD, median	0.036118	0.036042	same robust dispersion
p99 - p01, median	0.314048	0.313771	same dynamic range

3.2 Poison-box geometry and contrast

The 20 poison boxes are extremely small: median area is 0.0748% of the image and the maximum is 0.2132%. Median width and height are 34.7 and 27.2 pixels. Elongation is heterogeneous, from almost square to 7.21x, so a single aspect-ratio rule cannot isolate the poison. Median patch-to-local-background contrast is only +0.214 local standard deviations; one annotated patch is slightly darker than its neighborhood. The poison is therefore a weak, local concept embedded in a statistically ordinary image.



Poison area	Elongation	Local contrast	Image shift	Template density
median 0.075%	median 1.72x	median +0.214 sigma	mean differs <0.04%	1 detection median
range 0.016-0.213%	95th percentile 5.79x	one patch below background	unlearn vs 40 test images	634 / 2,000 empty

The poison occupies very little area and has weak contrast; global image statistics do not separate unlearn from test.

Figure 2. EDA of the complete unlearn set and poisoned submission template. The confidence histogram is a compact binning of the exact 2,593 scores; all reported quantiles are exact.

3.3 Consequence for validation

Because only 20 positive poison examples exist and global image distributions overlap, ordinary train/validation splitting is not informative. Box-level leave-one-out tests would have very high variance and no clean negative labels. Reliable local validation must instead test invariants: poison firing on held-out synthetic concepts, retention on deliberately non-poison elongated streaks, number of new boxes, IoU retention against the original model and output density. Our early diagnostics measured density and confidence mass, but only the later published solutions used concept-level synthetic clean/poison separation as a primary gate.

4. Fundamental detector

4.1 Architecture reconstruction

The checkpoint matches Detectron2's COCO-Detection/retinanet_R_50_FPN_3x configuration adapted to one class. A ResNet-50 bottom-up backbone produces multi-scale features; an FPN creates P3 through P7 with 256 channels. Two parallel RetinaNet heads each contain four 3 x 3 convolutions. The classification output has shape 7 x 256 x 3 x 3 because seven anchors are used per location for one class. The box output has shape 28 x 256 x 3 x 3 because each anchor predicts four deltas.

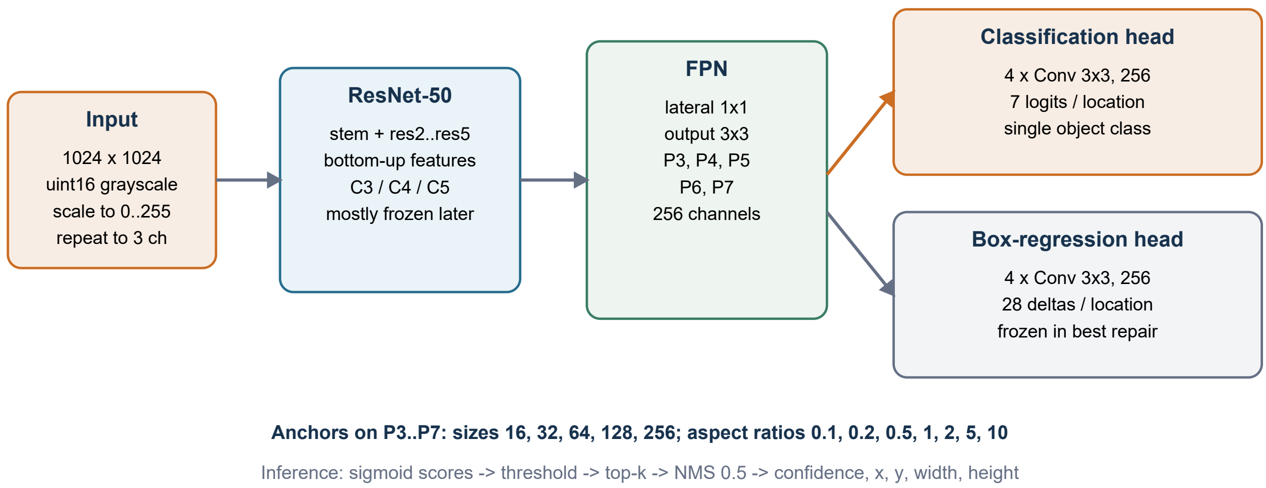


Figure 3. Fundamental detector and the modules later frozen or repaired. The elongated anchor ratios are essential for streak geometry.

4.2 Anchor configuration

Exact architectural compatibility was critical. The anchor sizes are 16, 32, 64, 128 and 256 pixels on P3..P7. Each level uses aspect ratios 0.1, 0.2, 0.5, 1, 2, 5 and 10. Using Detectron2 defaults would change the number and geometry of anchors, fail checkpoint shape compatibility or alter predictions. This was one of the first contract checks in the project.

4.3 Where the poisoned behavior can live

A backdoor can be distributed across the backbone, FPN and head, but the task evidence suggested a classification-dominant circuit. Empty-label training produced zero positive anchors and zero box-regression loss; nevertheless confidence and detection counts changed quickly. Pruning classification-subnet channels altered whether existing candidate boxes survived while 2,581 of 2,582 geometries remained exactly aligned. These observations do not prove that the backbone is clean, but they justify a minimal intervention in the classification path.

Module	Parameters / tensors	Best-private treatment	Reason
ResNet-50 backbone	stem + res2..res5	Frozen	avoid broad representation drift
FPN P3..P7	lateral/output layers	Frozen	preserve scale-space geometry
Classification subnet	4 conv layers	Pruned then trainable	highest poison selectivity
Classification score	7-anchor logits	Trainable	direct confidence control
Box subnet + predictor	4 conv + 28 deltas	Frozen	maCADD penalizes box drift
Total detector	36,253,667 params	2,376,455 trainable (6.55%)	minimal repair surface

5. Model-repair method

5.1 Activation-guided pruning

For each of the four classification-subnet convolution layers, forward hooks collect activations on all 20 unlearn images. Each annotated poison box is projected into the feature-map coordinates. ReLU activation inside poison regions is averaged per channel and compared with activation from size-matched random background patches. The channel poison score is the foreground-minus-background mean.

```
q_l,c = mean[ ReLU(a_l,c) | poison boxes ] - mean[ ReLU(a_l,c) | random background ]
```

The top 15% of channels in every classification convolution are zeroed, corresponding to 38 of 256 channels per layer and 152 channel outputs overall. The operation is surgical: it removes the most poison-selective directions without changing the FPN, box subnet or anchor geometry. A calibrated-noise branch was also tested, but noise did not establish a stable poison-only separator and was not part of the best private candidate.

5.2 Classifier-only empty-label fine-tuning

The pruned checkpoint becomes the initialization and EWC anchor. All 20 images are registered with empty annotations, which tells RetinaNet that the known poison response should not be classified as an object. Only parameters whose names contain cls_subnet or cls_score remain trainable. This gives 2,376,455 trainable parameters out of 36,253,667. The configuration is batch size 4, learning rate 2.5e-4, 10 iterations, no learning-rate steps, no augmentations and seed 42.

```
L(theta) = L_focal-empty(theta) + 250 * sum_j || theta_j - theta_pruned,j ||^2
```

The EWC term here is a diagonal-free L2 anchor rather than a Fisher-weighted classical EWC implementation. Its practical role is still important: it restricts the classifier to a small neighborhood of the post-pruning solution. Since all targets are empty, unrestricted optimization would monotonically suppress objectness; the short horizon and anchor penalty stop training before useful streak behavior collapses.

5.3 Metric-aware inference

The original poisoned model remains the high-recall candidate generator at confidence 0.05. The repaired model is an oracle for candidate survival, not the geometry source. For each original box i, the closest repaired box is found. If best IoU is below 0.5, s_diff(i)=1; otherwise s_diff(i)=1-s_repaired/s_original. A geometry likelihood s_geo is computed by fitting a Gaussian to log(width,height) of the 20 poison boxes and converting Mahalanobis distance to a similarity.

```
p_poison(i) = 0.90 * s_diff(i) + 0.10 * s_geo(i)
```

Original detections below 0.20 are removed. If p_poison >= 0.55, confidence is set to epsilon 0.01; if p_poison <= 0.25, original confidence is retained; between the thresholds it is linearly faded. Epsilon detections overlapping a strong box by IoU >= 0.20 are removed to avoid duplicates. This asymmetric remap is aligned with maCADD: retain geometry, lower confidence only where repaired-model evidence supports forgetting.

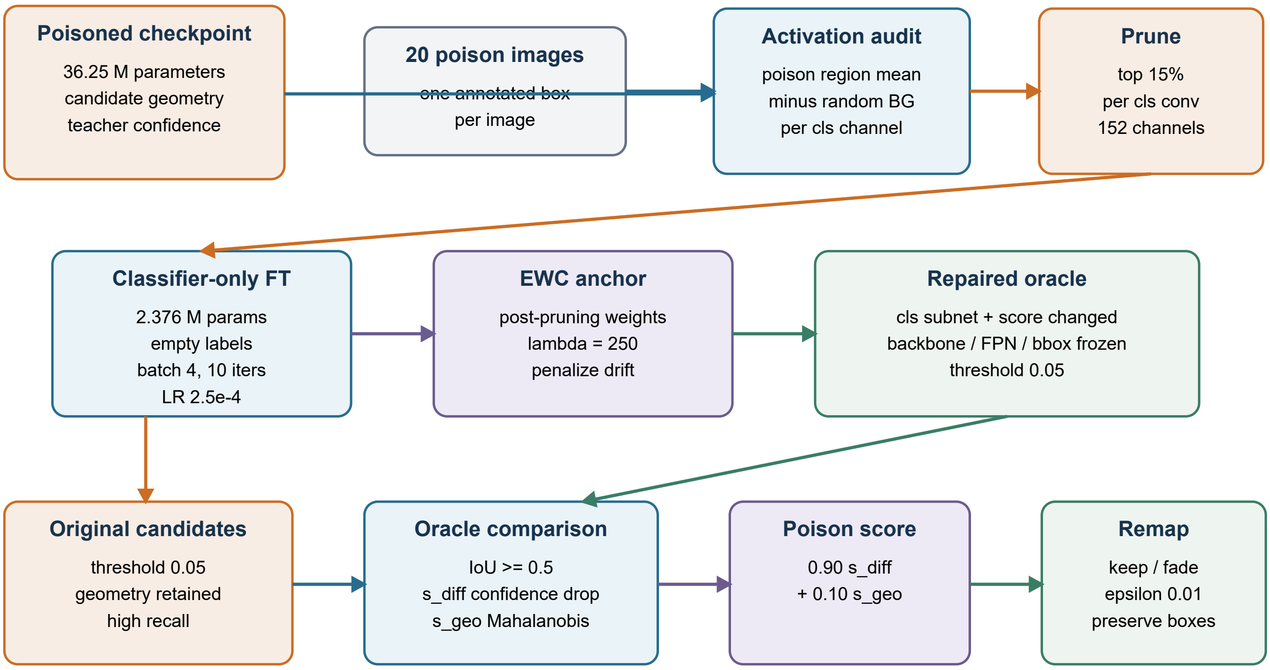
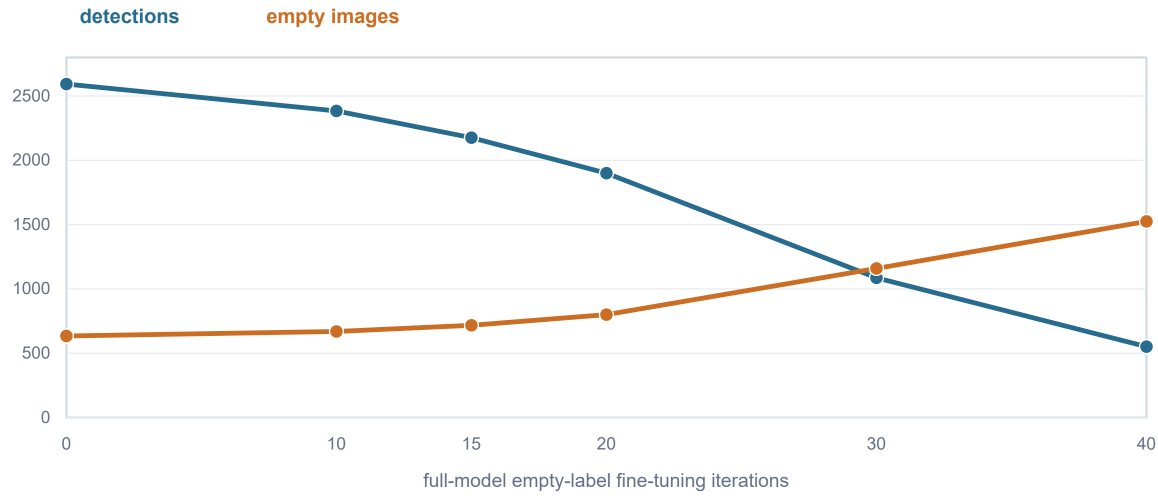


Figure 4. Complete best-private stack from poisoned checkpoint to submission. Blue/orange modules are changed; gray/green paths preserve the original detector.

6. Hyperparameter search and negative results

6.1 Why full-model fine-tuning fails

The strict baseline trained almost the entire detector (36.03 M trainable parameters) on the same 20 empty-label images with learning rate 1e-4 and batch size 4. Iteration count was the dominant hyperparameter. At 10 iterations, 2,384 detections remained; at 15, 2,176; by 30, only 1,086; and by 40, 551. Empty images rose from 634 in the original template to 1,525. The model was learning the trivial solution 'detect nothing' rather than a poison-selective boundary.



Beyond the narrow 10-20 iteration region, output density collapses faster than poison selectivity improves.

Figure 5. Output collapse under full-model empty-label fine-tuning. The 20-iteration detection count is interpolated from the submitted profile because its detailed audit was not retained.

Iterations	Trainable scope	Detections	Empty images	Public score	Decision
0	none	2,593	634	poisoned template	reference
10	36.03 M params	2,384	669	not submitted	plausible profile
15	36.03 M params	2,176	717	290.9630	regression
20	36.03 M params	about 1,900	about 800	259.7886	best strict baseline
30	36.03 M params	1,086	1,159	not submitted	over-suppressed
40	36.03 M params	551	1,525	not submitted	collapse

6.2 Pruning, noise and EWC

The Jayhawk branch ranked classification channels by poison-region activation, pruned 15%, injected score-scaled Gaussian noise of amplitude 0.01 and fine-tuned with EWC lambdas 3,000 to 6,000. It improved the public basin to 247.8306 but produced unstable detection density across operation order, pruning fraction and noise amplitude. The lesson was not that pruning is ineffective; rather, pruning plus broad FPN/head training and strong noise confounds poison removal with retention damage.

6.3 Synthetic causal pairs

A later no-submit experiment created original, masked, moved and perturbed variants for each poison image. It trained selected FPN and classification layers (3.295 M parameters) for 80 iterations at 5e-5 with outside-patch feature consistency weight 0.15. Training was numerically unstable: classification loss repeatedly spiked above 1 and reached 3.88. The output returned to 2,180 detections, 714 empty images and confidence mass 1,010.82, a known 257-style basin. The run was rejected because intervention alone did not provide a strong clean-streak preservation target.

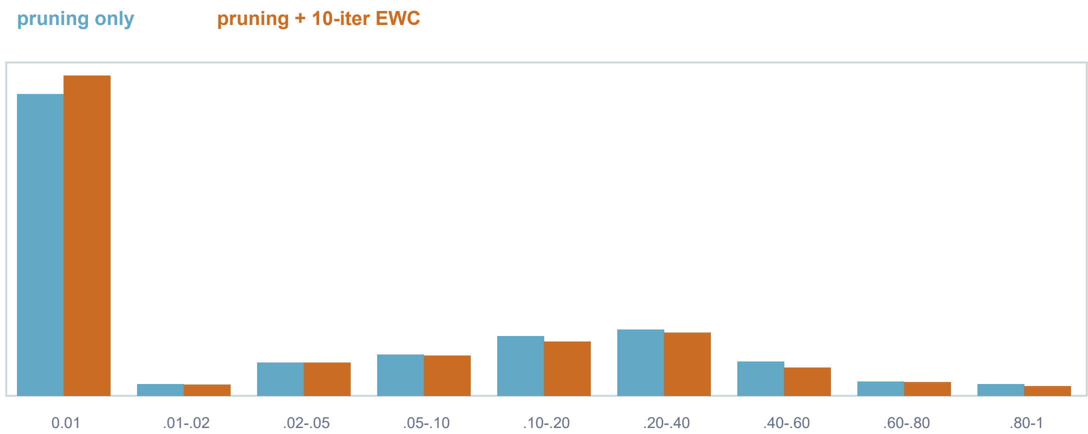
6.4 Why the best hyperparameters are coherent

The final configuration combines the useful part of each branch while removing the source of instability: classification-only scope instead of full-model updates; deterministic zeroing instead of noise; 10 iterations instead of 20-80; and a moderate anchor penalty of 250 rather than large EWC values applied to broader modules. These choices are not justified by an unbiased local validation score, which does not exist. They are justified by invariants: unchanged geometry, stable density, exact output audits and better private leaderboard behavior than the public-best mask.

7. Ablation of the best private notebook

7.1 Pruning-only versus pruning plus short fine-tuning

A dedicated causal ablation ran the same inference and confidence remap with the public pruned checkpoint, then with an additional 10-iteration classifier-only EWC fine-tune. Both variants produced exactly 2,582 detections and 634 empty images. Therefore the fine-tune did not win by altering output density or geometry. It changed which boxes received epsilon confidence.



The short fine-tune moves 83 additional detections to epsilon while preserving all 2,582 geometries.

Figure 6. Confidence-distribution ablation. Phase B changes decisions, not the detection set.

Variant	Detections	Empty	Confidence mass	Epsilon count	Decision agreement proxy
Pruning only	2,582	634	324.516	1,357	corr 0.101
Pruning + FT + EWC	2,582	634	295.557	1,440	corr 0.767
Original Sanidhya output	2,582	634	296.528	1,440	reference

The mixed candidate exactly reproduced all 2,582 epsilon/non-epsilon decisions of the public Sanidhya output and matched all 2,582 geometries within 0.02 pixels. Pruning alone differed on 82 epsilon decisions and had 9.4% higher confidence mass. This is strong evidence for a mixed mechanism: pruning exposes the poison-selective subspace; the short anchored fine-tune calibrates the classifier inside that subspace.

7.2 Leaderboard result

The submitted Phase B candidate scored 229.1934 public and 306.3502 private, the best private result on the account and final rank 89. The previous public reproduction scored 229.2314 / 306.3586, confirming that the improvement was small but reproducible. The result is far from the winner, but it is the most defensible model-based outcome in this study because the mechanism, checkpoint hash, output profile and private score agree.

7.3 Why it works

- It targets the classification circuit identified by poison-region activation instead of perturbing all 36 M parameters.
- It freezes the box regressor, so high-IoU geometry from the original model is preserved for maCADD matching.
- It uses the repaired network as an oracle over original candidates, avoiding new unmatched detections.
- The 10-iteration horizon and EWC anchor prevent the empty-label objective from converging to global suppression.
- Its confidence map follows the metric's asymmetry: suspicious detections are demoted to epsilon rather than deleted indiscriminately.

Interpretation: the notebook is good because it separates *where boxes are* from *whether they should be trusted*. Geometry remains the high-recall knowledge of the fundamental model; the repair acts primarily on classification confidence.

8. Forensic analysis of the public-best notebook

8.1 What the notebook actually executes

The notebook title suggests external RetinaNet fine-tuning, but the scored code does not load Detectron2 or train weights. It reads the organizer sample submission, contains a literal list of 450 global detection indices, sets every non-selected confidence to 0.02, and adds 0.15 to selected original confidences with a maximum of two selected boxes per image. It asserts an exact SHA-256 for the resulting CSV.

Property	Poisoned template	Public-best CSV
Rows	2,000	2,000
Detection geometries	2,593	same 2,593
Empty images	634	same 634
Selected boxes	not applicable	450
Non-selected confidence	original	0.02
Selected confidence	original	min(1, original + 0.15)
Confidence mass	1,401.460	399.506
Leaderboard	template reference	219.4257 public / 308.0971 private

8.2 Why public improved but private did not

The fixed mask encoded information accumulated from public feedback and prior oracle experiments. On the public split it selected a useful subset and reduced the cost of the remaining 2,143 boxes to epsilon. However, it did not learn a transferable image-to-detection rule. Its public gain of 9.77 points over Phase B became a private regression of 1.75 points. The split reversal is exactly what one expects when a decision boundary is represented by test-set indices rather than by features.

The correct claim is therefore: 219.4257 is the best public submission, not the best trained model. The 306.3502 Phase B candidate is the best private and scientifically strongest notebook. Treating the mask replay as a training result would overstate the method and obscure the central lesson about generalization.

9. Results and leaderboard context

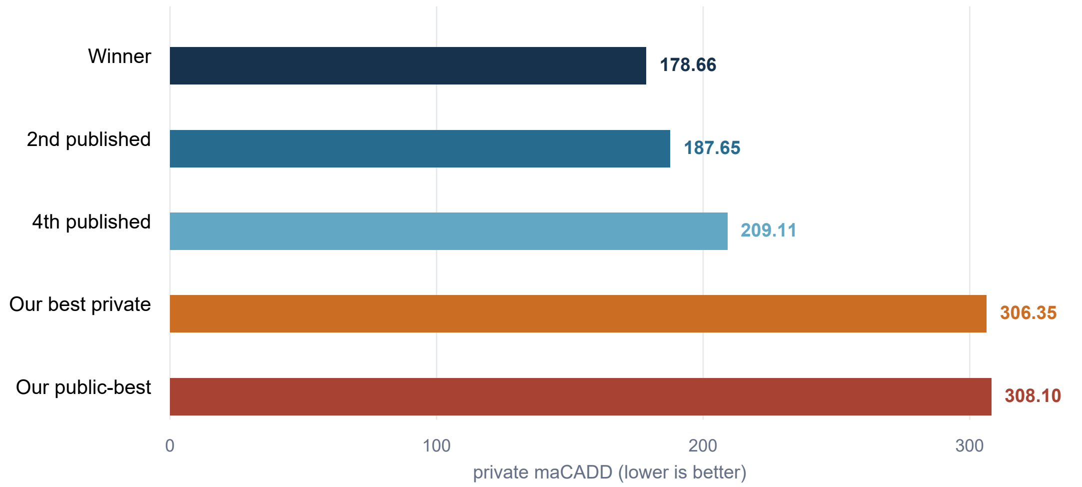


Figure 7. Final private comparison. The public-best mask is slightly worse privately than the model-based Phase B candidate.

Milestone	Public	Private	What changed
Initial repair-suppress	449.9513	609.1727	aggressive suppression collapse
Strict empty-label FT	259.7886	346.9767	full model, narrow iteration region
Preserve-soft B1	257.1246	342.1067	limited preservation losses
Public CV reproduction	245.3014	334.1291	better public, weak private
Jayhawk pruning + EWC	247.8306	324.2157	activation-guided circuit repair
Sanidhya Phase B	229.1934	306.3502	best private model
Contrastive trial v2	232.3060	319.7617	inferior selection precision
Fixed 450-box mask	219.4257	308.0971	best public only

The winner scored 178.6560 private, a 127.6942-point gap. The documented second-place method scored 187.6500 using dense teacher-student distillation: non-poison anchor logits and box deltas were pinned to the teacher, poison anchors were pushed toward a soft target, and held-out synthetic poison/clean concepts gated models before ensembling. Our method shares the preserve/forget decomposition but applies it more coarsely through pruning and output-level oracle comparison.

10. What remains missing

10.1 Dense anchor-level preservation

Freezing most modules prevents broad drift, but it cannot preserve the classification logits of every non-poison anchor. The leading published solution directly distills classification logits and box deltas outside a poison-anchor mask. This gives a local constraint at every FPN location and is stronger than global EWC anchoring.

10.2 A mask aligned with anchors

The poison boxes are small and sometimes highly elongated. IoU between a small streak and a containing anchor can be low. Intersection-over-anchor or center-in-box masking better identifies which classification units express the poison. Our activation projection approximated this boundary at feature-map level but did not supervise the final anchor logits explicitly.

10.3 Synthetic clean contrast

Moved and masked poison patches teach invariance, but they do not define what a real clean streak must retain. A stronger suite should contain two separated families: bright beaded or comet-like poison concepts to forget and smooth elongated clean streaks to preserve. Held-out concept firing, not training loss, should determine model selection.

10.4 Confidence estimation after stable set selection

The public-best experiment optimized selection and calibration jointly through a fixed mask. A transferable system should first form a stable consensus detection set across genuinely diverse repaired models, then restore confidence from the teacher only where multiple models agree. Calibration cannot rescue a contaminated or split-specific set.

10.5 Reproducibility boundary

The public repository intentionally excludes competition images, checkpoints, notebook caches and submissions. The paper reports SHA-256 values, output profiles and exact configurations where stored. The lightweight train.py and infer.py in the repository form a tested release component, but they do not claim to reproduce the Detectron2 Kaggle pipeline or the final leaderboard entry by themselves.

11. Threats to validity

The clean model and labels remain unavailable, so mechanism claims rely on intervention, output invariants and leaderboard outcomes. The 20 poison examples are too few for conventional statistical generalization. The 40-image test EDA sample is systematic rather than random and is used only to detect gross distribution shift. Several decisions were influenced by public scores, making public comparisons adaptive. The winner did not publish a detailed method by the audit date; comparisons therefore focus on documented second- and fourth-place solutions.

The ablation identifies the role of the 10-iteration classifier update relative to pruning, but it does not independently score pruning-only on the leaderboard. Its conclusion is based on exact output comparison and the scored mixed candidate. Similarly, the interpolated 20-iteration density in Figure 5 is labeled approximate because its detailed audit was not retained.

12. Conclusion

The Neural Debris study progressed from indiscriminate suppression to a narrow, evidence-based repair of the detector's classification circuit. EDA showed why this was difficult: poison patches occupy less than one tenth of one percent of an image at the median, have weak local contrast and are embedded in backgrounds statistically indistinguishable from test images. Full-model empty-label training therefore learns global suppression long before it learns a trustworthy poison concept.

The best private notebook succeeds by limiting degrees of freedom. It prunes poison-selective classification channels, updates only 6.55% of parameters for 10 iterations, anchors them to post-pruning weights, freezes geometry and remaps original candidate confidence from repaired-model survival. The resulting 306.3502 private score is not close to the winner, but its mechanism and ablation are coherent. The lower 219.4257 public score came from a fixed index mask and should not be interpreted as a better trained model.

The next technically meaningful experiment is a dense anchor-level teacher-student repair with intersection-over-anchor poison masks and held-out synthetic clean/poison concept validation. That would combine the strongest insight of this project - preserve geometry and constrain the repair surface - with the representation-level supervision used by the best published solutions.

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Appendix A. Exact best-private configuration

Component	Value
Base config	COCO-Detection/retinanet_R_50_FPN_3x.yaml
Checkpoint SHA-256	f6c21faa2a5b...a5a6f0e657c
Image transform	uint16 / 65535 -> x255 -> repeat grayscale to RGB
Anchors	sizes 16/32/64/128/256; ratios 0.1/0.2/0.5/1/2/5/10
Pruning	top 15% FG-minus-BG channels in each cls conv
Trainable	head.cls_subnet + head.cls_score only; 2,376,455 params
Training	batch 4; SGD; LR 2.5e-4; 10 iterations; seed 42; no augmentation
Regularization	L2 EWC-style anchor to post-pruning classifier; lambda 250
Candidate/oracle threshold	0.05 / 0.05
Oracle match	best IoU >= 0.5
Poison probability	0.90 s_diff + 0.10 s_geo
Remap	MIN_KEEP .20; P_LO .25; P_HI .55; EPS .01
Output	2,582 detections; 634 empty; confidence mass 295.557; 1,440 epsilon
Scores	229.1934 public; 306.3502 private; rank 89

Appendix B. Complete scored submission history

#	Date	Description	Public	Private
1	Jun 08	repair-suppress-001	449.9513	609.1727
2	Jun 11	simple fine-tune baseline	259.7886	346.9767
3	Jun 11	simple fine-tune MAX_ITER=15	290.9630	389.8185
4	Jun 17	r02 cluster probe	259.2333	344.6891
5	Jun 18	interpolation beta 0.05	257.6416	343.6334
6	Jun 18	interpolation beta 0.02	259.4834	344.7010
7	Jun 19	triangular interpolation	258.8906	344.3229
8	Jun 19	interpolation beta 0.06	257.6576	343.3147
9	Jun 21	preserve-soft B1	257.1246	342.1067
10	Jun 22	fusion consensus average	257.7674	342.4261
11	Jun 23	public pruning + fine-tuning	247.6207	327.7771
12	Jun 23	public CV de-poisoning	245.3014	334.1291
13	Jun 24	Jayhawk reproduction	247.8306	324.2157
14	Jun 26	Jayhawk diagnostic re-submit	247.8306	324.2157
15	Jun 28	noise 0.03	248.5938	326.7423
16	Jul 06	NDR trial 1 / Sanidhya	229.2314	306.3586
17	Jul 18	Sanidhya Phase B	229.1934	306.3502
18	Jul 21	NDR trial v2	232.3060	319.7617
19	Jul 23	fixed confidence mask	219.4257	308.0971